

Robotix - Inventory Turnover Analysis

Q3 2024 Report

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Executive Summary

Our Q3 2024 inventory turnover analysis shows a ratio of 3.5x, meeting the industry benchmark but leaving room for improvement. Total inventory value stands at \$7.2M, with Days Inventory Outstanding (DIO) at 104.3 days. Working capital tied to inventory represents 18% of total current assets.

Key Metrics Summary:

Overall Turnover Ratio: 3.5x
DIO: 104.3 days
Total Inventory Value: \$7.2M
Quarter-over-Quarter Change: -5%

Inventory Turnover by Category

Product Category	Turnover Ratio	Avg. Inventory Value	DOI (Days)
Industrial Robots	4.2x	\$2.8M	86.9
Collaborative Robots	3.8x	\$1.9M	96.1
Mobile Robots	2.9x	\$1.6M	125.9
Components & Accessories	4.5x	\$0.9M	81.1

Detailed Inventory Analysis

Raw Materials

Total Value: \$2.1M
Turnover: 4.8x
Major Components:

- Motors: \$620K
- Controllers: \$485K
- Sensors: \$395K
- Frames: \$600K

Work-in-Progress (WIP)

Total Value: \$1.8M
Average Assembly Time: 12 days
Aging Profile:
- 0-30 days: 65%
- 31-60 days: 25%
- >60 days: 10%

Finished Goods

Total Value: \$3.3M
Product Line Distribution:
- RX-2000 Series: \$1.2M
- CollabBot Series: \$0.9M
- MobileNav Series: \$0.8M
- Accessories: \$0.4M

Location Analysis

Facility	Inventory Value	Turnover	Utilization
Minneapolis Mfg	\$4.8M	3.8x	82%
Portland DC	\$2.4M	3.1x	75%

Quarter-over-Quarter Comparison

Metric	Q2 2024	Q3 2024	Change
Turnover Ratio	3.7x	3.5x	-5.4%
Total Inventory	\$6.9M	\$7.2M	+4.3%
DIO	98.6	104.3	+5.7 days

Cash Flow Impact

Working Capital Analysis: - Inventory carrying cost: \$216K/quarter (3% of value) - Cash conversion cycle: 156 days - DSO: 45 days - DPO: 38 days

Risk Assessment

High-Risk Areas: 1. RX-2000 component obsolescence (\$180K exposure) 2. CollabBot sensor inventory (tech upgrade pending) 3. Supply chain concentration in Asian markets

Recommendations

1. Implement JIT for standard components - Potential working capital release: \$400K - Target implementation: Q4 2024
2. Optimize slow-moving inventory - Bundle older models with service contracts - Target clearance: \$250K by year-end
3. Enhanced demand forecasting - Implement ML-based forecasting system - Expected accuracy improvement: 15%

Action Items

Immediate Actions: 1. Review \$480K slow-moving inventory for clearance 2. Negotiate VMI agreements with top 3 suppliers 3. Implement cycle counting program in Portland DC 4. Accelerate RX-2000 component consumption

90-Day Objectives: - Reduce DIO to 95 days - Increase turnover ratio to 3.8x - Release \$500K working capital - Implement automated reorder points

Next Steps

1. Weekly review of slow-moving inventory
2. Monthly cross-functional inventory optimization meetings
3. Quarterly supplier performance reviews
4. Implementation of new inventory management software

This analysis identifies significant opportunities for inventory optimization while maintaining service levels. Key focus areas include reducing slow-moving inventory and implementing more sophisticated demand forecasting tools.

Note: All figures in USD unless otherwise stated.

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